Ana Lorena Ortiz Loyola

Frankfurt am Main, Germany
+49 176 62145797 • a01750283@tec.mx • LinkedIn

Data science student (Tecnológico de Monterrey, expected 2027) with hands-on experience building and validating Python/SQL data pipelines for a large-scale reporting platform at Deutsche Bundesbank (AnaCredit). Contributed a significant, independently-owned component to a complex data project (lattice-based cryptography research) and can clearly communicate its technical contribution. Quick to ramp up on new platforms such as Palantir Foundry, motivated by LLM-powered applications and data/AI platform engineering.

Core Competencies

- **Programming**: Python (Pandas, NumPy, Scikit-Learn), SQL; building and extending backend data interfaces and pipelines
- **Data & AI Platform Engineering Support**: Building and validating pipelines for a large-scale reporting platform, quick to onboard onto new platforms such as Palantir Foundry
- **Data Quality & Validation**: Automated migration tests (row counts, referential integrity, aggregate consistency), deviation and anomaly detection, rule-based validation
- **Machine Learning & LLM-Adjacent Work**: K-Means clustering, Random Forest, SVM; motivated by LLM-powered application development
- **Agentic AI Tools**: Regular use of Claude Code as an AI coding agent to accelerate data-tool development and automation
- **Collaboration & Documentation**: Clear technical communication of project contributions, Jira, Git, structured technical documentation

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** — **Frankfurt am Main, Germany**
  Data Engineer / Intern — *04/2026–07/2026*
  - Built SQL/Python data transforms to prepare and validate backend reporting data, developing automated migration tests (row counts, referential integrity, aggregate consistency) before roll-out
  - Implemented automated error detection identifying deviations (duplicate IDs, value-range violations) in reporting data, then structured findings for stakeholder review
  - Automated Jira ticket creation and emails to reporting banks, translating technical findings into clear, actionable communication for non-technical stakeholders
  - Significantly reduced response time to data quality issues through end-to-end ownership, from detection to resolution

Projects

- **Tecnológico de Monterrey**
  Predictive Modeling – Healthcare Optimization — *10/2024–12/2024*
  - Analyzed 25+ years of Mexican national health data to identify patient risk profiles and resource bottlenecks
  - Built predictive models (Random Forest, Neural Networks) and K-Means clustering to segment high-risk groups, directly analogous to clustering structured records into meaningful categories

- **Personal Project (GitHub)**
  Banking Data Pipeline – Data Cleaning & Audit Log *05/2026*
  - Built an end-to-end pipeline for simulated bank transactions (150 accounts) with a normalized SQLite database and foreign-key integrity checks
  - Staged data cleaning with a fully auditable log; automated balance maintenance via SQL triggers; built using **Claude Code** as an AI coding assistant to accelerate implementation
- **Tecnológico de Monterrey**
  Lattice-Based Cryptography Research *10/2024–02/2025*
  - Implemented and analyzed the complexity of lattice-based cryptographic algorithms (LWE, NTRU) in Python
  - Contributed to a scientific publication analyzing security parameters and runtime behavior across lattice dimensions

Education

- **Tecnológico de Monterrey** **Mexico**
  BSc in Data Science *2022–2027*
  Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: data analysis & statistics, machine learning & AI, cryptography & security, NLP, numerical optimization.
- **Universität Göttingen** **Germany**
  Exchange Semester *2025–2026*
- **Tecnológico de Monterrey** **Mexico**
  Abitur *2019–2022*
  Grade: 1.0. National Physics Olympiad 2020; Honorable Mention, Physics Olympiad 2021.

Languages

- Spanish (native), English (C1, TOEFL iBT 105/120), German (C1).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglisse, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.